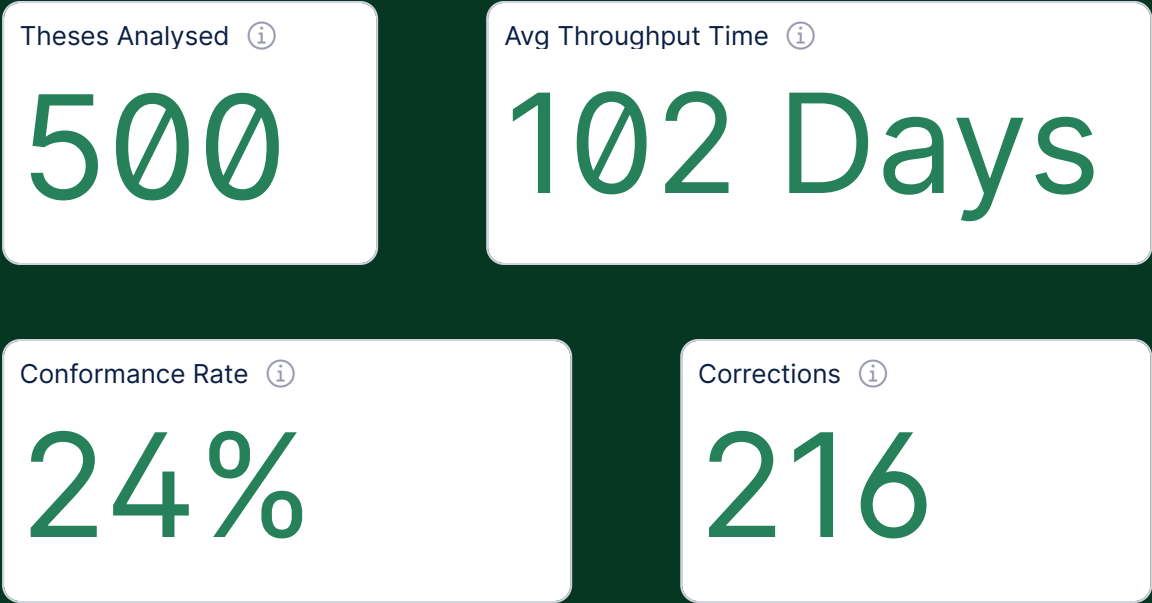


MASTER THESIS PROCESS INTELLIGENCE DASHBOARD

TITM UNIVERSITY



A PROJECT BY:
HENRY LEWIS



TITM
UNIVERSITY

Total Variants

41

Average Throughput In Days

102

Theses With Corrections

216

Only **23.8%** of theses follow the ideal process path

Correction loops are the primary driver of process deviation

41 distinct process variants identified across **500** cases



Variant Explorer

cases covered

24%1 of 41 variant119 of 500 cases

Expand Controls ↗



Activities

24 of 24

Throughput Time Distribution

Event Log ⓘ

Event Log Event Log Activity

Default

▼

from

Please select an event

▼

First

▼

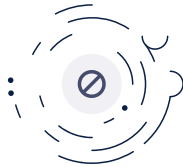
to

Please select an event

▼

Last

▼



Please select events in the "From Event" and "To Event" selectors to view the throughput time histogram

46

Rejected Submissions

10

Failed Defenses

135

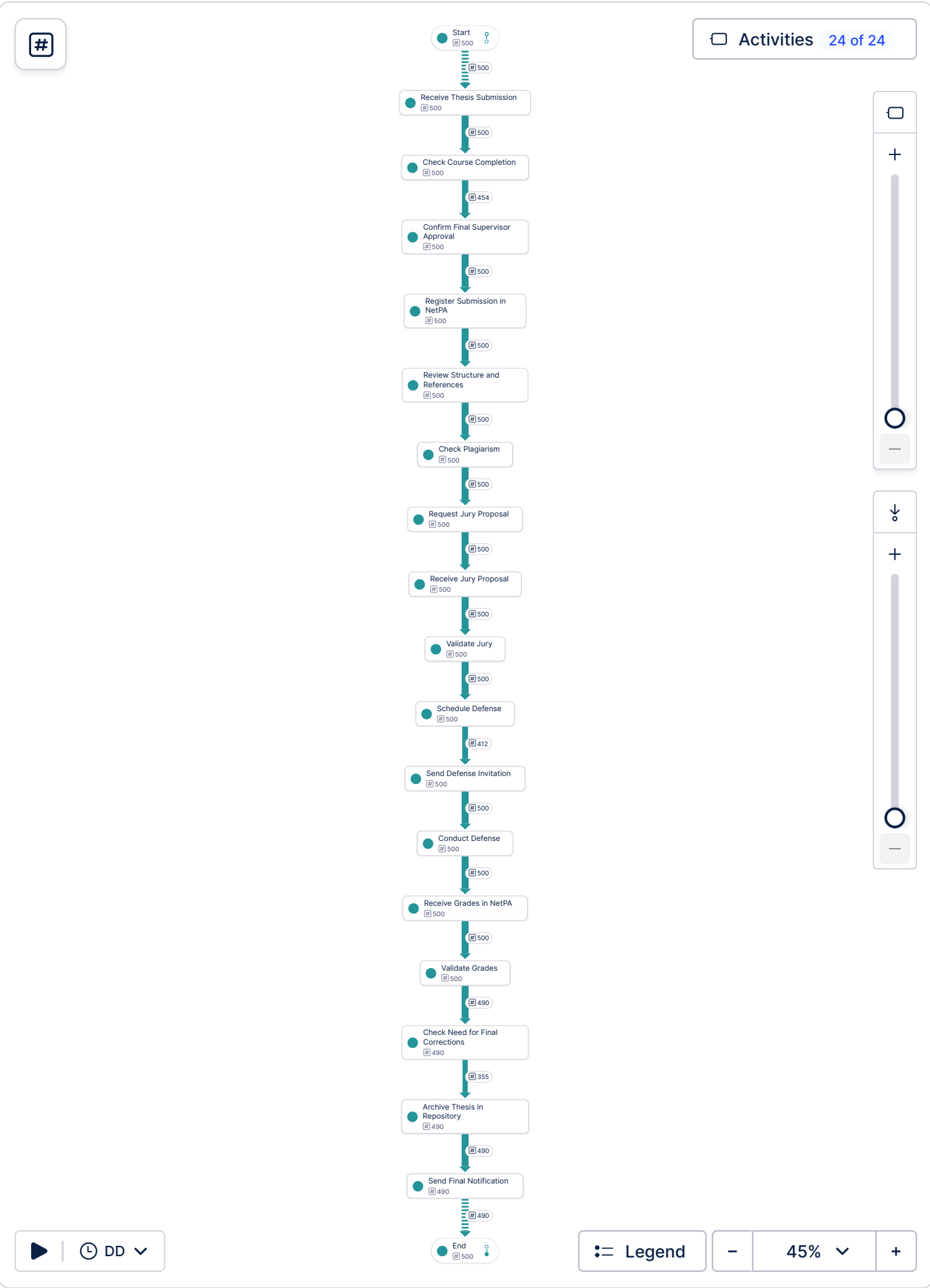
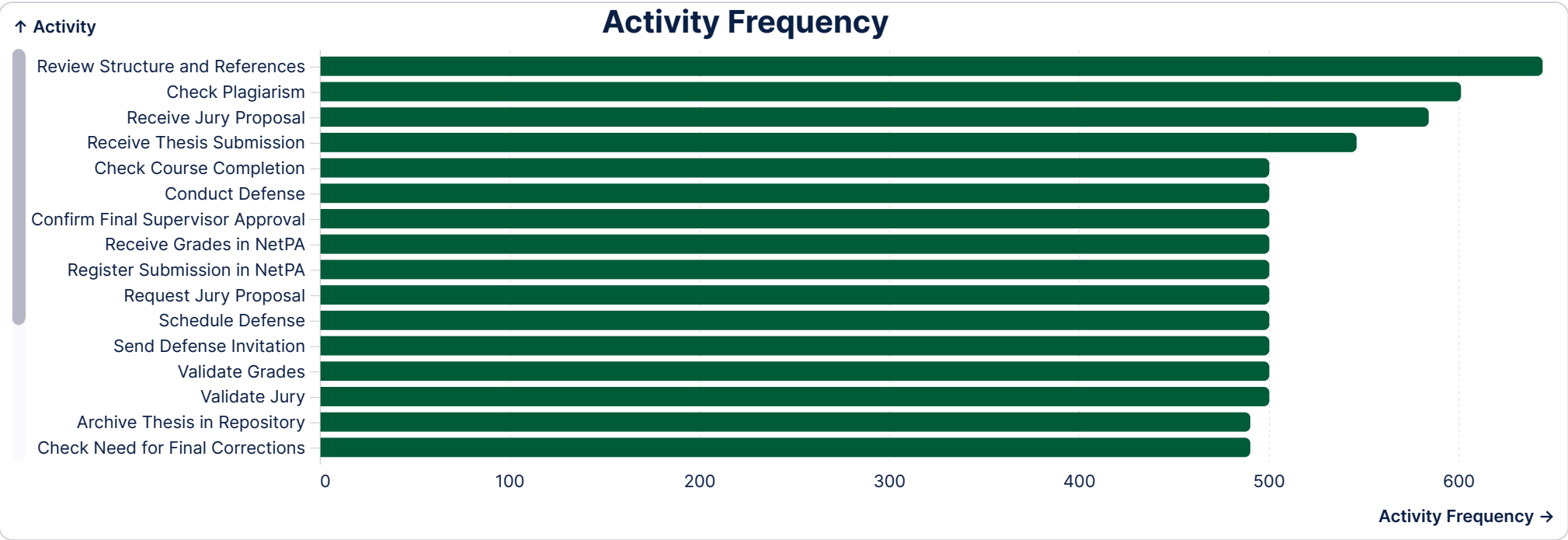
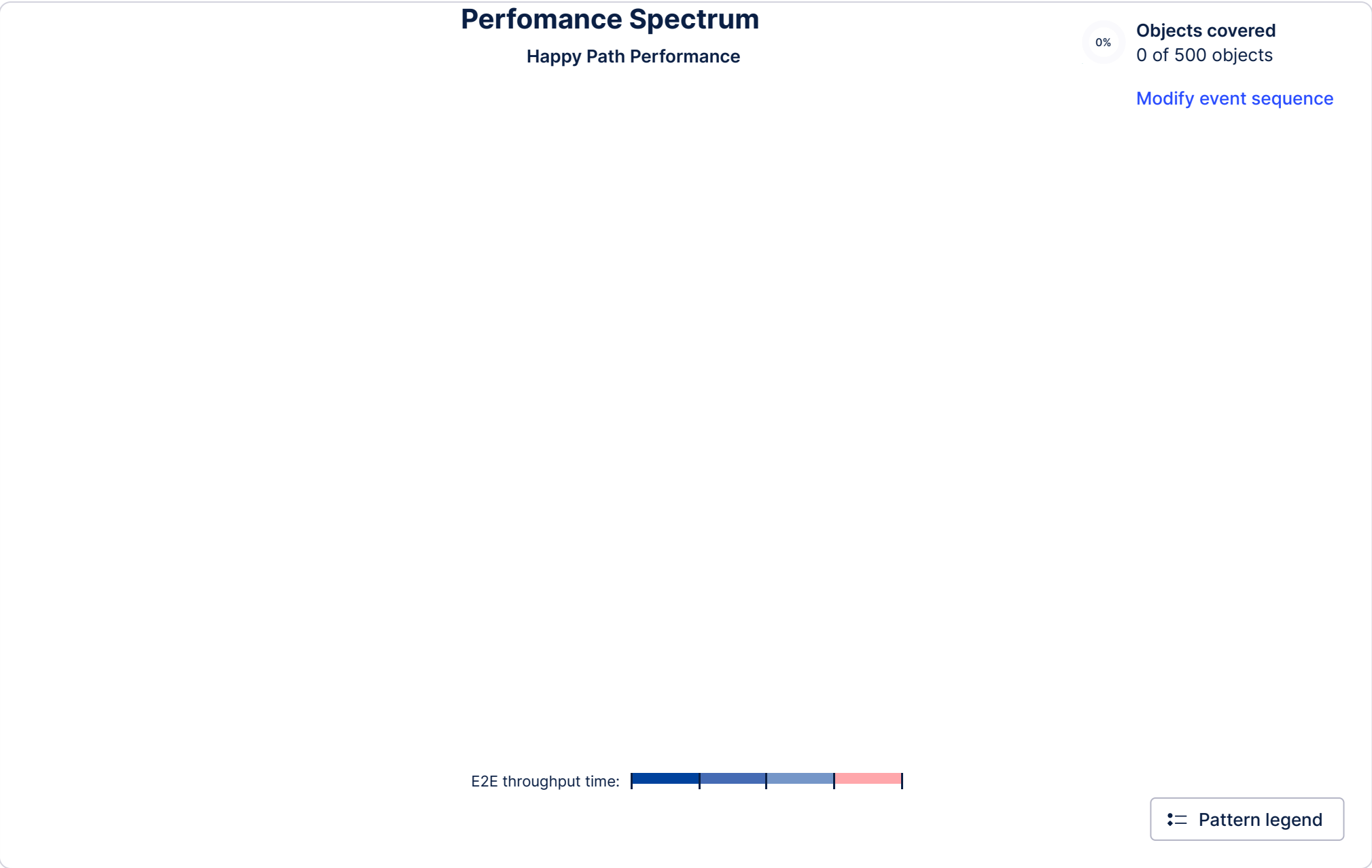
Final Corrections Required

88

Rescheduled Defences

84

Jury Revisions



119

Happy Path Cases ⓘ

381

Cases With Deviations ⓘ

24%

Conformance Rate ⓘ

216

Corrections ⓘ

Conformance rate

Event Log Event Log Activity 23.80%

Deviations (14)

Search

Request Thesis Corrections

Unexpected event

Event Log Event Log Activity 216 (43.20%)

TPT impact +4 hours

Receive Revised Thesis

Unexpected event

Event Log Event Log Activity 216 (43.20%)

TPT impact +4 hours

Review Structure and References

Occurred too often

Event Log Event Log Activity 144 (28.80%)

TPT impact -16 minutes

Receive Final Corrected Version

Unexpected event

Event Log Event Log Activity 135 (27.00%)

TPT impact +4 days

Check Plagiarism

Occurred too often

Event Log Event Log Activity 101 (20.20%)

TPT impact +9 hours

Reschedule Defense

Unexpected event

Event Log Event Log Activity 88 (17.60%)

TPT impact +4 days

Request Jury Revision

Unexpected event

Event Log Event Log Activity 84 (16.80%)

TPT impact +3 days

Receive Jury Proposal

Occurred too often

Event Log Event Log Activity 84 (16.80%)

TPT impact +3 days

Receive Thesis Submission

Occurred too often

Event Log Event Log Activity 45 (9.20%)

TPT impact +8 hours

#

Q

Event Log Event Log Activity

Receive Thesis Submission

Check Course Completion

Confirm Final Supervisor Approval

Register Submission in NetRA

Review Structure and References

Check Plagiarism

Request Jury Proposal

Receive Jury Proposal

Validate Jury

Schedule Defense

Send Defense Invitation

Conduct Defense

Receive Grades in NetRA

Validate Grades

Check Need for Final Corrections

Archive Thesis in Repository

Send Final Notification

Legend

20%

Happy Path Conformance

23.8% Follows Happy Path

76.2% Contains Deviations

Activity

Explicit Deviation Activities by Cases

Receive Revised Thesis

Request Thesis Corrections

Receive Final Corrected Version

Reschedule Defense

Request Jury Revision

Reject Submission

Record Failed Defense

0

50

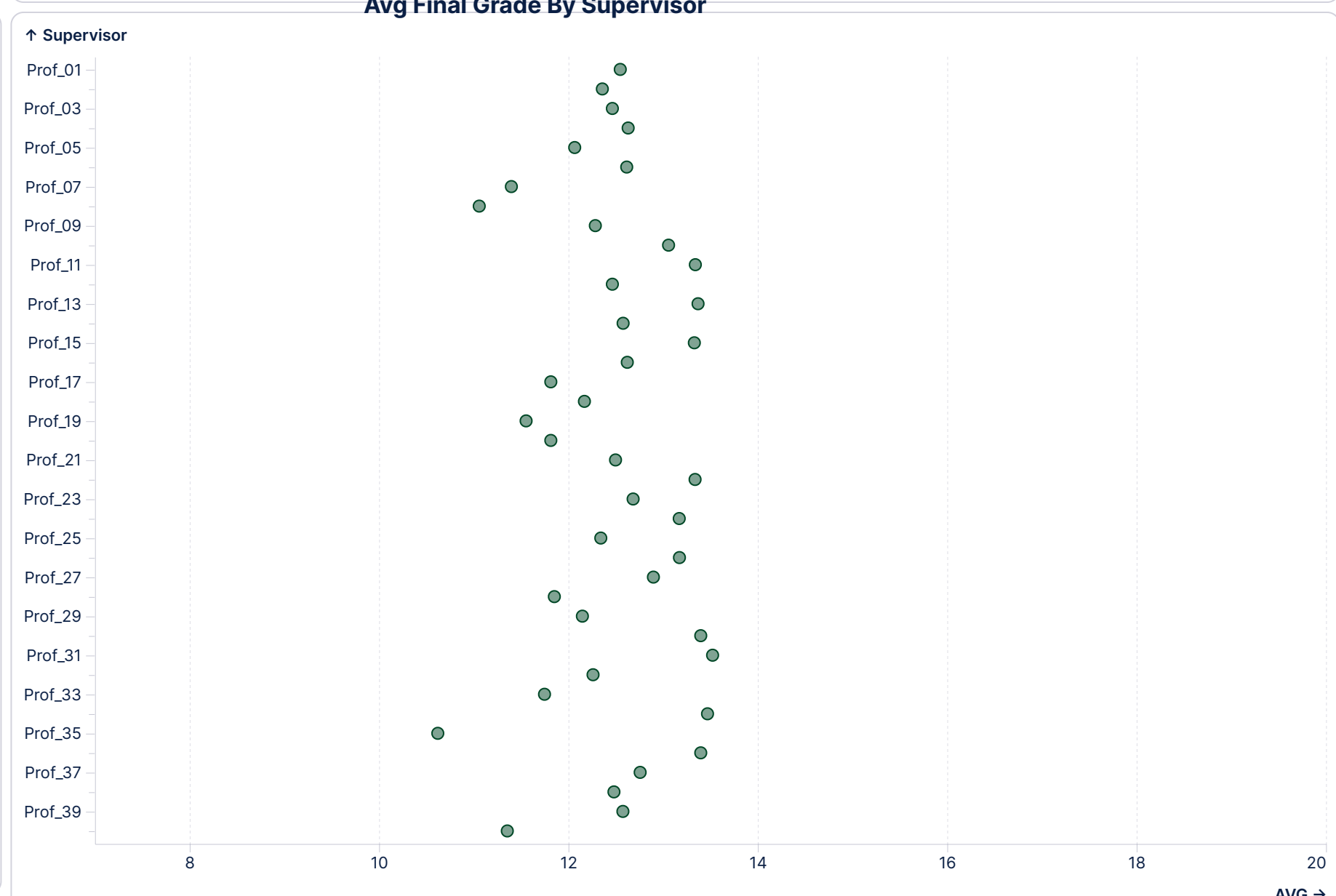
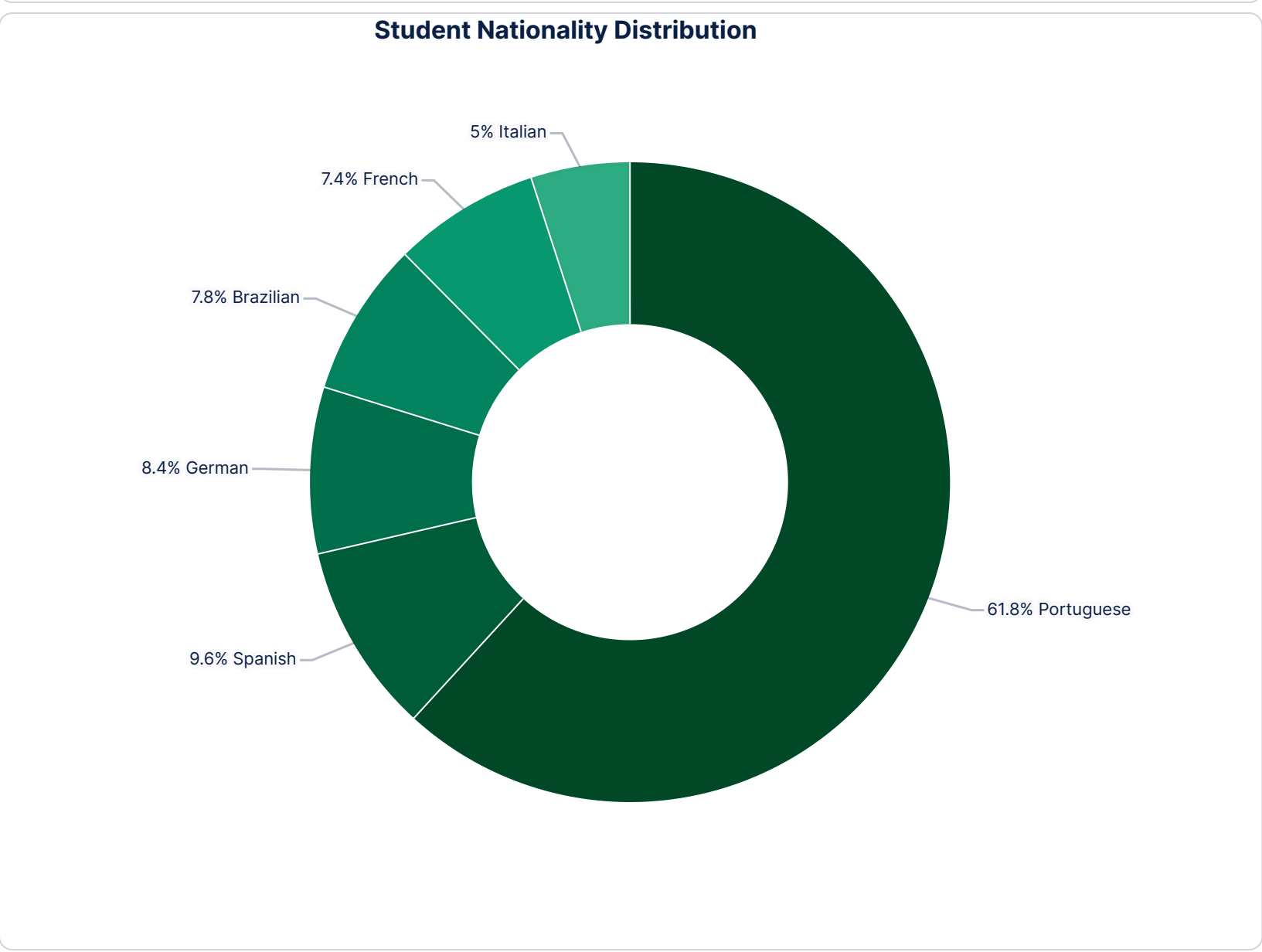
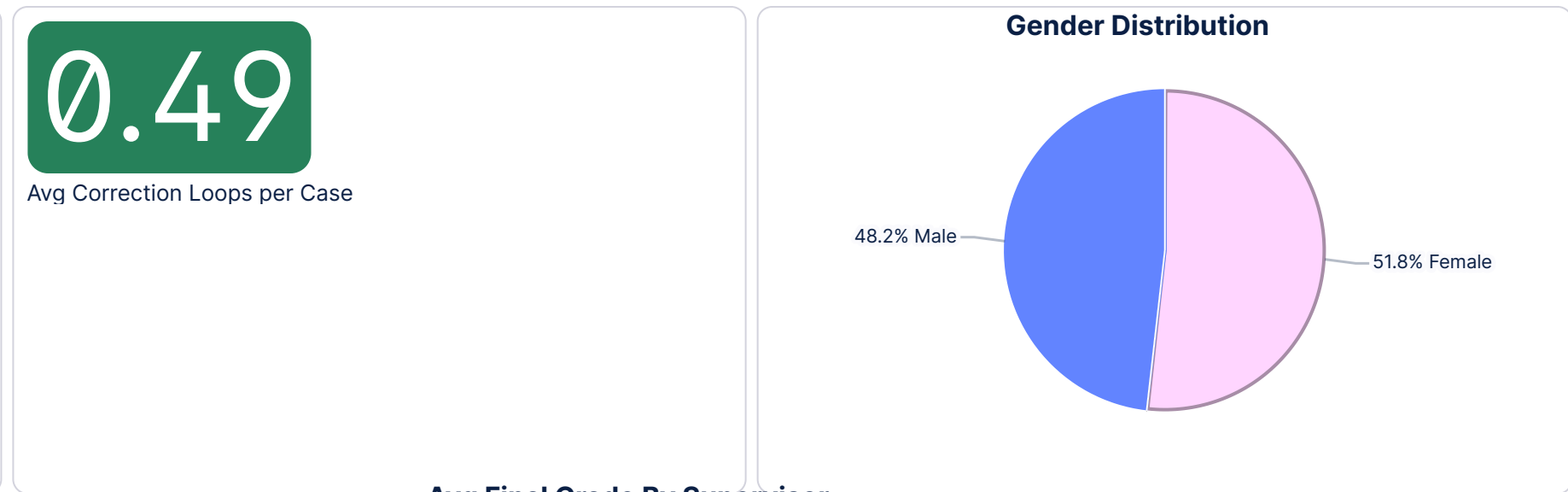
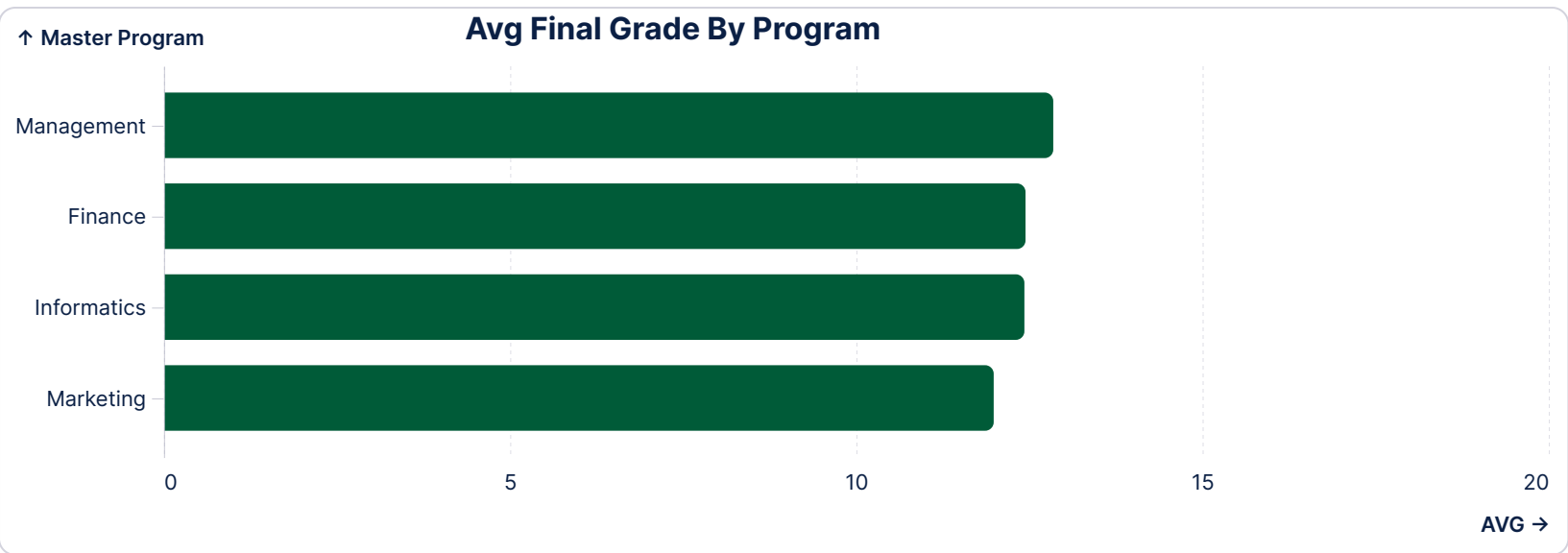
100

150

200

Cases →

Case Id	Student Number	Master Program	Supervisor	Nationality	Gender	Age	Final Grade	Correction Loops	
T0001	100,001	Management	Prof_03	French	Male	38	10.00	2	
T0002	100,002	Management	Prof_35	Portuguese	Female	30	14.10	0	
T0003	100,003	Management	Prof_11	Portuguese	Female	54	15.00	1	
T0004	100,004	Marketing	Prof_16	Portuguese	Female	41	10.00	0	
T0005	100,005	Finance	Prof_25	German	Male	34	15.00	1	
T0006	100,006	Marketing	Prof_05	Spanish	Male	49	9.60	1	
T0007	100,007	Management	Prof_07	Portuguese	Male	50	11.30	0	
T0008	100,008	Informatics	Prof_14	German	Female	34	16.00	0	



H1: Supervisor correction variation

- Correction loops vary strongly by supervisor.
- Suggests supervision practices influence rework.

✓

H2: Defense rescheduling drivers

- Reschedules vary across student groups.
- Jury availability is likely not the only driver.

✗

H3: Informatics correction rate

- Informatics is not the highest group.
- Finance has more post-defense corrections.

✗

H4: Correction loop persistence

- Initial corrections slightly increase final correction risk.
- Effect exists, but remains weak.

✓

H5: Demographic performance effect

- Grades are similar across gender groups.
- No clear age / performance pattern appears.

✗

H6: Supervision practice effect

- Issues cluster around certain supervisors.
- Stable grades suggest a process issue.

✓

↑ Supervisor

Prof_01
Prof_03
Prof_05
Prof_07
Prof_09
Prof_11
Prof_13
Prof_15
Prof_17
Prof_19
Prof_21
Prof_23
Prof_25
Prof_27
Prof_29
Prof_31
Prof_33
Prof_35
Prof_37
Prof_39

Supervisor Correction Variation

Avg Correction Loops →

↑ Supervisor

Prof_01
Prof_03
Prof_05
Prof_07
Prof_09
Prof_11
Prof_13
Prof_15
Prof_17
Prof_19
Prof_21
Prof_23
Prof_25
Prof_27
Prof_29
Prof_31
Prof_33
Prof_35
Prof_37
Prof_39

Supervisor Reschedule Variation

AVG Reschedules →

↑ Gender

Female
Male

Average Grade By Gender

Final Grade →

Gender: Female Male

Average Reschedules By Nationality

26.16% Brazilian
19.88% German
17.55% French
17.39% Spanish
15.31% Portuguese
3.71% Italian

↑ Master Program

Finance
Informatics
Management
Marketing

Post Defence Corrections By Program

Post-defense corrections by programme →

Initial Corrections → Final Corrections

52.76%
47.24%

Yes No

Corrections ⓘ

216

Rescheduled Defenses

88

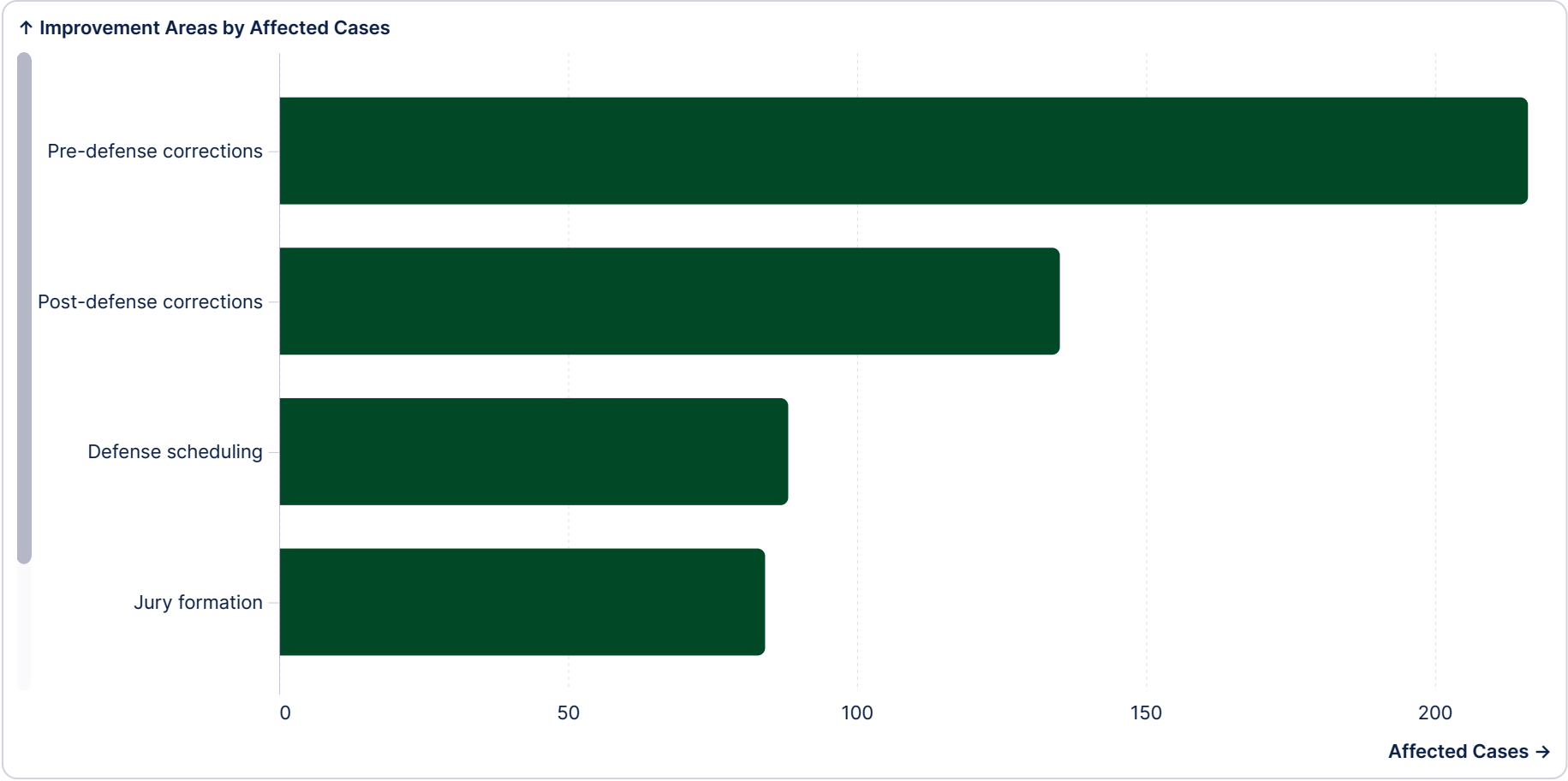
Jury Revisions

84

Final Corrections Required

135

Live Case Risk Monitoring							⌵
Filter by Risk Flag to prioritise intervention							
Case Id	Supervisor	Programme	Correction Loops	Defence Rescheduled	Final Correction	Risk Flag	⋮
T0001	Prof_03	Management	2	Yes	Yes	High Correction Risk	⌵
T0002	Prof_35	Management	0	No	No	On Track	
T0003	Prof_11	Management	1	No	No	On Track	
T0004	Prof_16	Marketing	0	No	No	On Track	
T0005	Prof_25	Finance	1	No	Yes	Final Correction Risk	
T0006	Prof_05	Marketing	1	No	No	On Track	
T0007	Prof_07	Management	0	Yes	No	Scheduling Risk	
T0008	Prof_14	Informatics	0	No	No	On Track	
T0009	Prof_11	Management	0	No	No	On Track	
T0010	Prof_36	Marketing	0	Yes	Yes	Scheduling Risk	
T0011	Prof_18	Management	0	No	No	On Track	
T0012	Prof_16	Marketing	1	Yes	No	Scheduling Risk	
T0013	Prof_35	Informatics	1	No	Yes	Final Correction Risk	
T0014	Prof_09	Finance	0	Yes	No	Scheduling Risk	
T0015	Prof_08	Management	0	No	No	On Track	



Prioritised Improvement Actions

Recommended actions based on thesis process mining findings

1

Targeted Supervisor Development

Standardise supervision practices and reduce repeated corection loops.

2

Pre-Submission Quality Checklist

Prevent avoidable structure, reference and plagiarism corrections before submission.

3

Earlier Jury Formation

Start jury planning earlier to reduce revisions and defense rescheduling pressure.

4

Real-Time Risk Monitoring

Flag cases with correction loops, reschedules or final corrections for early intervention.

5

International Scheduling Support

Provide extra scheduling support where reschedule rates are higher.

Management conclusion

The main improvement opportunity is reducing avoidable process variation through better supervision standards, earlier jury planning, pre-submission quality checks and live case monitoring.